

Festival Shopping Behavior Analysis

A comprehensive analysis of 3,900 customer transactions to inform strategic business decisions.





PROJECT OVERVIEW

Project Overview

Dataset Scale

3,900 purchases analyzed
across multiple product
categories

Analysis Scope

Customer demographics,
purchase behavior, product
preferences, and subscription
patterns

Objective

Deliver actionable insights to guide strategic business decisions



DATA FOUNDATION

Dataset Overview

Data Structure

- 3,900 rows × 18 columns
- Customer demographics tracked
- Purchase details captured
- Shopping behavior analyzed

Key Features

- Age, gender, location data
- Product categories & sizes
- Payment & shipping methods
- Review ratings collected

Data Preparation & Cleaning

01

Data Loading

The raw dataset was programmatically ingested leveraging the pandas library, establishing the foundational structure for subsequent analytical processing.

02

Initial Exploration

Comprehensive initial data exploration was performed to discern data schema, integrity constraints, and to compute descriptive statistics for preliminary insights.

03

Missing Data Handling

Missing values within the 'review_ratings' feature (n=37) were systematically addressed through median imputation to preserve data integrity.

04

Feature Engineering

Advanced feature engineering techniques were applied to derive new attributes, such as granular age demographics and cumulative purchase frequency metrics, enhancing predictive capability.

05

Database Integration

The meticulously prepared and cleaned dataset was subsequently persisted within the PostgreSQL relational database management system for secure storage and efficient retrieval.



FINANCIAL ANALYSIS

Revenue Analysis by Customer Segment

\$59.7.. \$157K \$75K

Average Purchase Value

Represents the mean monetary value derived from each individual customer transaction.

Male Customer Revenue

Total revenue generated from the male customer segment, indicating a substantial contribution.

Female Customer Revenue

Aggregate revenue generated specifically from the female customer segment.

Analysis of Shipping Method on ATV

The average transaction value (ATV) categorized by shipping preference:

- Express Shipping: \$60.48
- Standard Shipping: \$58.46

Revenue Contribution by Subscription Status

Total revenue disaggregated by customer subscription engagement:

- Non-subscribers: \$170,000
- Subscribers: \$63,000

Top-Rated Products

1

Gloves

Attained the highest average customer rating of 3.86, indicating superior product reception.

2

Sandals

Achieved an average rating of 3.84, reflecting robust customer satisfaction and product quality.

3

Boots

Maintained a consistent quality performance with an average rating of 3.82, signifying reliable customer approval.

4

Hat

Exhibited a strong market appeal as an accessory with an average rating of 3.80, demonstrating favorable customer perception.

5

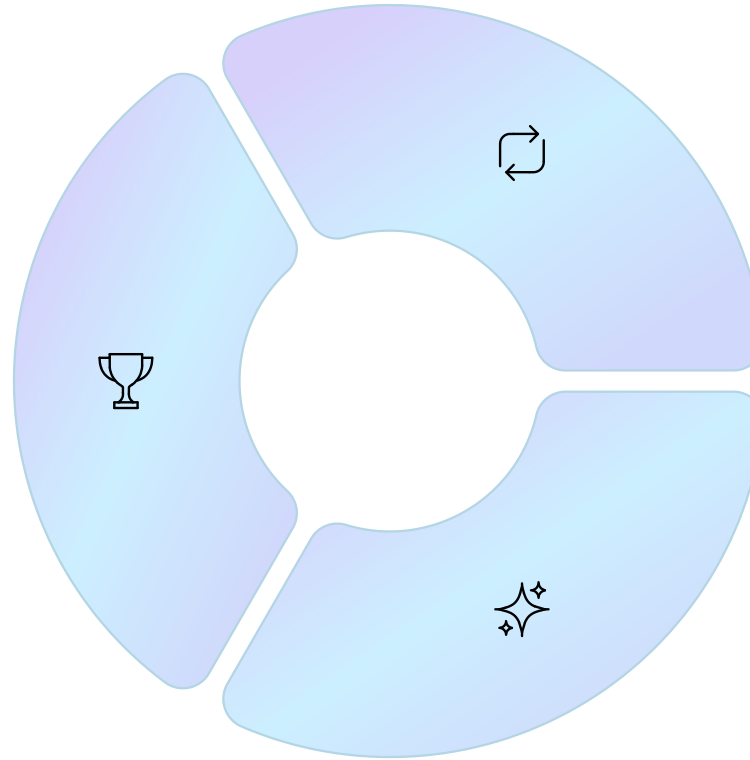
Skirt

Identified as a prominent apparel item, securing an average rating of 3.79.

Customer Segmentation

Loyal Customers

This segment comprises 3,116 individuals who exhibit consistent brand loyalty, representing a cornerstone of our customer base.

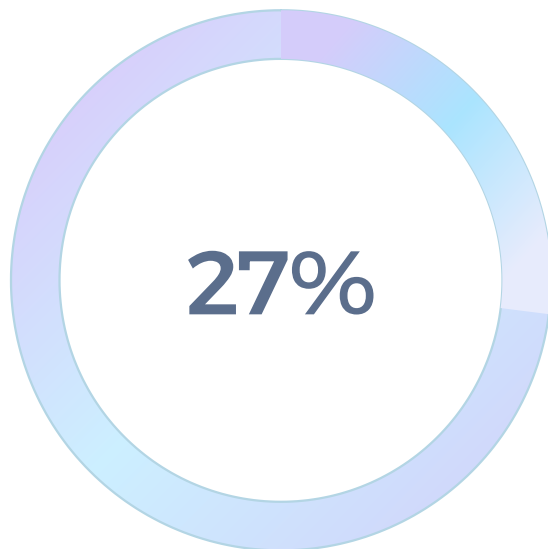


Returning Buyers

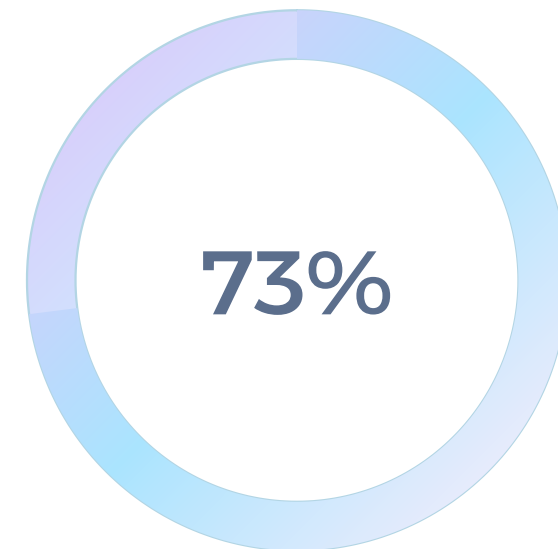
Consisting of 701 customers, this segment demonstrates established repurchase patterns, indicating successful retention efforts and potential for increased lifetime value.

New Shoppers

This emerging segment includes 83 new customers, signifying recent market penetration and presenting a primary focus for targeted acquisition strategies.

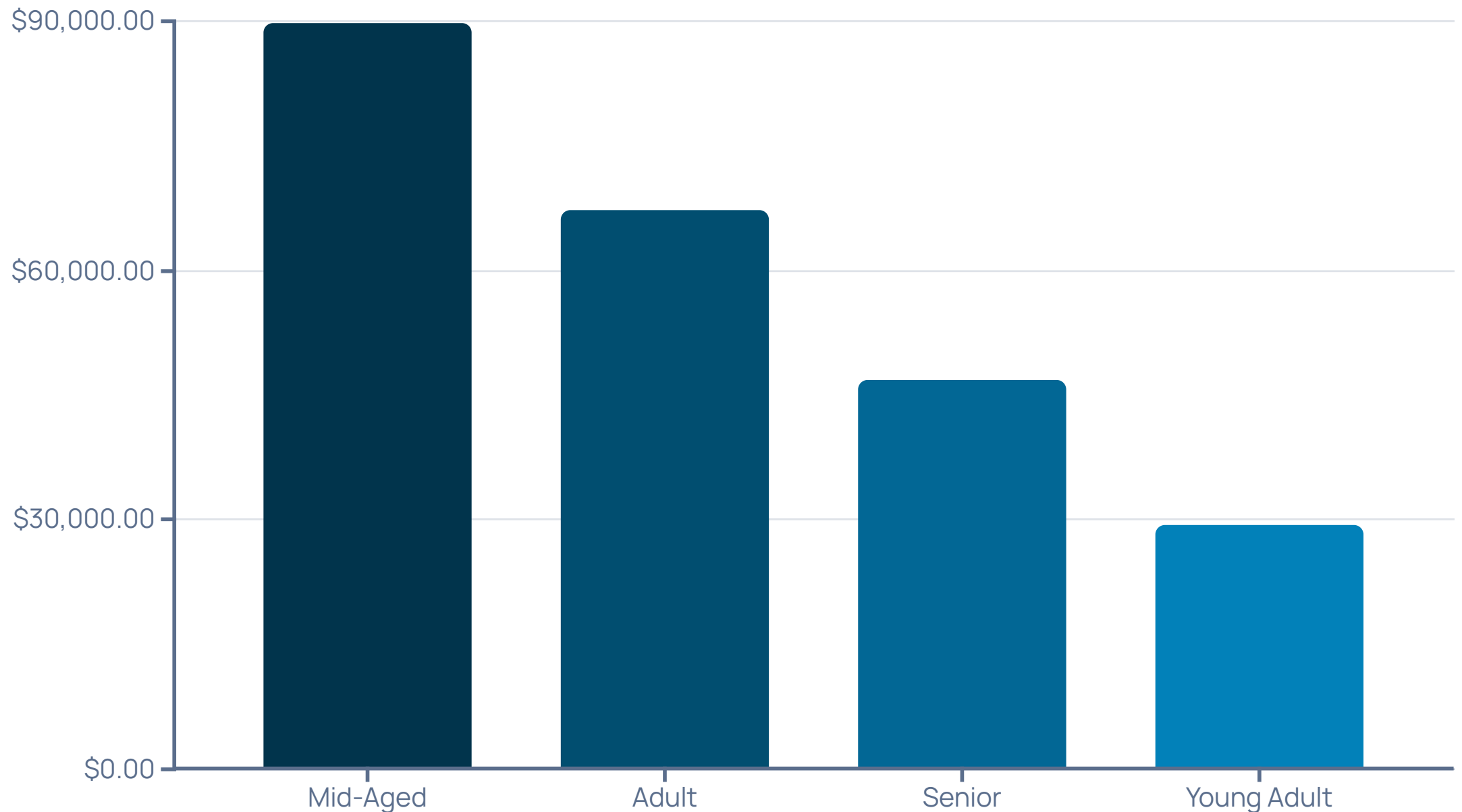


Subscribed Customers



Non-Subscribed Customers

Revenue Distribution by Age Group



Analysis and Strategic Implications

Young adults drive the highest revenue, followed by middle-aged customers. Marketing efforts should prioritize these high-value demographic segments.

Discount Dependency Analysis



50%

Hat

Exhibits the highest discount rate, indicating strong price elasticity.



49.7%

Sneakers

Approximately 49.7% of sales involve discounts, suggesting a significant reliance on promotions.



49.1%

Coat

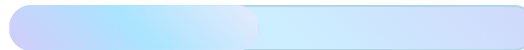
Displays a substantial dependency on discounts to drive sales volume.



48.2%

Sweater

Frequently featured in promotional campaigns to stimulate demand.



47.4%

Pants

Consistently utilizes discounting strategies to maintain sales performance.

Certain product categories show high discount dependency. Strategic pricing and promotional policies should balance revenue growth with margin preservation.

Business Recommendations



Boost Subscription Conversion

Initiate targeted campaigns to convert the 73% non-subscriber segment by offering exclusive benefits and value propositions.



Enhance Loyalty Programs

Develop and continuously optimize comprehensive loyalty programs to foster repeat patronage and expand the customer base.



Refine Targeted Marketing

Refine marketing strategies to precisely target mid-aged customer demographics and optimize engagement with users of express shipping services.



Optimize Discount Policy

Conduct a comprehensive strategic review of existing discount policies to achieve an optimal balance between sales volume expansion and sustained profit margins.



Strategic Product Positioning

Elevate top-rated products through strategic positioning within marketing campaigns to maximize perceived value and drive demand.



Forge Payment Partnerships

Establish strategic partnerships with leading digital payment providers to facilitate exclusive promotional offers during peak seasonal and festive periods.